Primal-Dual Optimization and Duality

DA5000W Linear Algebra and Optimization

December 16, 2025

End Sem

* least squares
* Orthogonality
* Gram Schmidt
* Unconstrained
  1st & 2nd cond
* Constrained
  KKT
  Duality
* Linear Program.

A general constrained optimization problem (*primal*):

LP simplex

constraint set

$$\min_{x\in\mathbb{R}^n} \quad f(x) \quad \text{Obj}$$
$$\text{s.t.} \quad g_i(x) \le 0, \quad i = 1, \dots, m, \quad \text{inequality}$$
$$h_j(x) = 0, \quad j = 1, \dots, p. \quad \text{equality}$$

- $f(x)$: objective function
- $g_i(x)$: inequality constraints
- $h_j(x)$: equality constraints

Introduce Lagrange multipliers:

- $\lambda_i \ge 0$ for inequality constraints
- $\mu_j \in \mathbb{R}$ for equality constraints

The **Lagrangian** is λ, μ dual variables

$$\Rightarrow \quad \mathcal{L}(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{j=1}^{p} \mu_j h_j(x).$$

Dual Optimization Problem

$q(\lambda, \mu) = \min_{x} L(x, \lambda, \mu)$

The **dual function** is defined as

$$q(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \mu).$$

$\nabla_x L(x, \lambda, \mu) = 0$

x in terms of λ

It's always concave in (λ, ν) and provides a lower bound on the primal optimal value

$q(\lambda, \mu)$

The **dual problem** is

$$(D) \quad \max_{\lambda,\mu} \; q(\lambda, \mu) \quad \text{s.t.} \quad \lambda \geq 0.$$

max $\Rightarrow$ min

*) $q(\lambda, \mu)$ concave fn

- Maximization of a concave function
- Often easier than the primal

Duality as Upper Bounds

$c = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

Consider the linear program:

$(1 \;\; 1)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$$\max_{x_1, x_2, x_3 \in \mathbb{R}} \quad \underline{x_1 + x_2} = f(x_1, x_2)$$
$$\text{s.t.} \quad \left.\begin{aligned} 2x_1 + x_2 + 4x_3 &\le 3, \\ x_1 + x_2 - 3x_3 &\le 1. \end{aligned}\right\} \text{constraint}$$

$\underbrace{\begin{pmatrix} 2 & 1 & 4 \\ 1 & 1 & -3 \end{pmatrix}}_{A}\underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{x} \underbrace{\le}_{\le} \underbrace{\begin{pmatrix} 3 \\ 1 \end{pmatrix}}_{b}$

Instead of solving the LP, we aim to prove lower and upper bounds on the optimal objective value.

$(3 \;\; 1)\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 3u_1 + u_2$

$\begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 4 & -3 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \ge \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \ge 0$

This is primal LP of the form

$\min \; b^T u$
s.t. $A^T u \ge c$
$u \ge 0$

$$\left\{ \begin{aligned} &\max \; \underline{c^T x} \\ &\text{s.t.} \quad \underline{Ax \le b}, \end{aligned} \right.$$

Let $u \in \mathbb{R}^m$ with $u \geq 0$. dual variable

Multiplying the constraints by u gives: $\Rightarrow Ax \leq b$

c^T $u^T Ax \leq u^T b$

$$u^\top Ax \leq u^\top b.$$

eliminate x
replace it w dual variable u

If we require

$u^T A \geq c^T$

$$A^\top u \geq c,$$

then

$u^T Ax \geq c^T x$

$$c^\top x \leq u^\top Ax \leq u^\top b.$$

Thus $b^\top u$ is an upper bound on the primal objective for all feasible x. To obtain the *best* upper bound, we minimize over all such u:

$c^T x \leq u^T b$

primal dual

$$\begin{aligned} \min_{u \in \mathbb{R}^m} \quad & b^\top u \\ \text{s.t.} \quad & A^\top u \geq c \\ & u \geq 0. \end{aligned}$$

primal min
$d^* \leq p^*$

$p^* \leq d^*$ for primal max

Converting a primal to a dual problem

The dual is obtained by:

- Introduce one dual variable per constraint
- Minimize the weighted sum of right-hand sides
- Ensure the weighted constraints dominate the objective

Introduce dual variables:

$$u_1 \geq 0 \quad \text{for constraint 1,} \qquad u_2 \geq 0 \quad \text{for constraint 2.}$$

We minimize the weighted sum of the right-hand sides:

$$\min\ 3u_1 + u_2.$$

$b^T u$

This gives one condition per primal variable:

$A^T u \geq c$

$$\begin{cases} 2u_1 + u_2 \geq 1 & \text{(coefficient of } x_1\text{)} \\ u_1 + u_2 \geq 1 & \text{(coefficient of } x_2\text{)} \\ 4u_1 - 3u_2 \geq 0 & \text{(coefficient of } x_3\text{)} \end{cases}$$

Dual Linear Program

Thus, the Dual LP can be formulated as,

x_1, x_2, x_3

u_1, u_2

D

$$\begin{aligned} \min_{u_1,u_2} \quad & 3u_1 + u_2 \\ \text{s.t.} \quad & 2u_1 + u_2 \geq 1, \\ & u_1 + u_2 \geq 1, \\ & 4u_1 - 3u_2 \geq 0, \\ & u_1, u_2 \geq 0. \end{aligned}$$

This is equivalent to writing,

(D)

$$\begin{aligned} \min_{u \geq 0} \quad & b^\top u \\ \text{s.t.} \quad & A^\top u \geq c. \end{aligned}$$

As the dual problem, using matrices for a general case.

Primal and Dual Linear Programs

Primal (P)

$$\Rightarrow \max_{x\in\mathbb{R}^n} \quad c^\top x \qquad \text{s.t.} \quad Ax \le b, \quad x \in \mathbb{R}^n$$

Decision variables: x
Feasible region: $Ax \le b$

$$\min \quad c^\top x \qquad \text{s.t.} \quad Ax \ge b$$

Dual (D)

$$\Rightarrow \min_{u\in\mathbb{R}^m} \quad b^\top u \qquad \text{s.t.} \quad A^\top u \ge c, \quad u \ge 0$$

Dual variables: u

$$\max \quad b^\top u \qquad \text{s.t.} \quad A^\top u \le c, \quad \underline{u \ge 0}$$

Primal–Dual Correspondence Table

$x_1 + x_2 = 2$

Maximization problem	Minimization problem
$\leq$ constraint	variable ≥ 0
$=$ constraint	unconstrained variable
$\geq$ constraint	variable ≤ 0
variable ≥ 0	$\geq$ constraint
unconstrained variable	$=$ constraint
variable ≤ 0	$\leq$ constraint

dual var
no sign constraint

Each constraint in the primal determines the sign of the corresponding dual variable, and each variable sign in the primal determines the type of constraint in the dual.

Why Is the Dual Often Easier to Solve?

$\rightarrow p^* \leq d^* \rightarrow$ weak duality

For a convex optimization problem, (primal - min, dual - max)

$\underset{\text{max}}{p^*} \leq \underset{\text{min}}{d^*}$

- Dual has fewer variables (2 instead of 3)
- Constraints are simpler to interpret
- Any feasible dual solution gives an immediate upper bound
- No need to find primal feasibility
- Primal feasibility $\Rightarrow$ lower bounds
- Dual feasibility $\Rightarrow$ upper bounds
- Dual $=$ best way to combine constraints
- Solving the dual can certify optimality without solving the primal

$\underset{\text{max}}{d^*} \leq \underset{\text{min}}{p^*}$

duality gap = 0

$p^* = d^* \rightarrow$ strong duality

Primal–Dual Linear Program

Primal (P):

$$\left\{\begin{aligned} \max_{x,y\geq 0} \quad & 2x+3y \\ \text{s.t.} \quad & -x+y\leq 3 \\ & x-2y\leq 2 \\ & x+y\leq 7 \end{aligned}\right\}$$

We solve the primal using the **simplex method**.
Introduce slack variables $s_1, s_2, s_3 \geq 0$:

$$\left\{\begin{aligned} -x+y+s_1 &= 3 \\ x-2y+s_2 &= 2 \\ x+y+s_3 &= 7 \end{aligned}\right\}$$

$s_1 = 3$
$s_2 = 2$
$s_3 = 7$

Objective:

$$\max z = 2x+3y$$

Initial Simplex Tableau

$[A \mid I]$

$\downarrow$

$[I \mid A^{-1}]$

Initial basic feasible solution:

$z =$

$$x = y = 0, \quad s_1 = 3,\ s_2 = 2,\ s_3 = 7$$

	x	y	s_1	s_2	s_3	RHS
s_1	-1	1	1	0	0	3
s_2	1	-2	0	1	0	2
s_3	1	1	0	0	1	7
z	-2	-3	0	0	0	0

Entering variable: y (most negative coefficient).
Minimum ratio test (positive entries in y column):

$$\frac{3}{1} = 3, \quad \frac{7}{1} = 7$$

Pivot element: (s_1, y).

Tableau After Pivot

	x	y	s_1	s_2	s_3	RHS
y	-1	1	1	0	0	3
s_2	-1	0	2	1	0	8
s_3	2	0	-1	0	1	4
z	-5	0	3	0	0	9

Entering variable: x.
Minimum ratio test (positive entries in x column):

$$\frac{4}{2} = 2$$

Pivot element: (s_3, x).

Final Simplex Tableau

$[I \mid A^{-1}]$

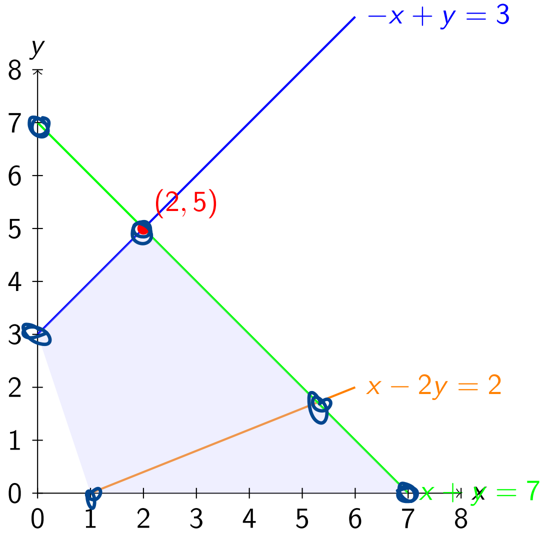
	x	y	s_1	s_2	s_3	RHS
y	0	1	$\frac{1}{2}$	0	$\frac{1}{2}$	5
s_2	0	0	$\frac{3}{2}$	1	$\frac{1}{2}$	10
x	1	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	2
z	0	0	$\frac{1}{2}$	0	$\frac{5}{2}$	19

All reduced costs are nonnegative $\to$ optimal.
The optimal solution is,

$$x^* = 2, \quad y^* = 5$$

$$z^* = 2x + 3y = 2(2) + 3(5) = 19$$

Graphical Solution of the Primal LP



KKT Conditions

(λ, μ) dual

For convex problems, x^*, λ^*, μ^* are optimal iff:

$g_i(x^*) \leq 0$

$h_j(x^*) = 0$

1. **Primal feasibility:** $g_i(x^*) \leq 0$, $h_j(x^*) = 0$
2. **Dual feasibility:** $\lambda_i^* \geq 0$
3. **Complementary slackness:**

$q(\lambda, \mu) = \max_x L(x, \lambda, \mu)$

$$\lambda_i^* g_i(x^*) = 0$$

$(D) \min_\lambda q(\lambda, \mu)$

4. **Stationarity:**

$$\nabla f(x^*) + \sum_i \lambda_i^* \nabla g_i(x^*) + \sum_j \nu_j^* \nabla h_j(x^*) = 0$$

Example: Simple Inequality Constraint

Primal problem:

$g(x) = 1 - x \leq 0$

$p^* = 1$

$$(P) \quad \min_x x^2 \quad \text{s.t.} \quad x \geq 1.$$

Rewrite constraint as $g(x) = 1 - x \leq 0$.

$f(x) + \lambda g(x)$

Lagrangian:

$$\mathcal{L}(x, \lambda) = x^2 + \lambda(1 - x), \quad \lambda \geq 0.$$

$x = \lambda/2$

$2x - \lambda = 0$

Dual function:

$d^* \leq p^*$

$$\Rightarrow \quad q(\lambda) = \inf_x (x^2 + \lambda(1 - x)).$$

$\min_x \mathcal{L}(x, \lambda)$

Minimizing over x gives $x = \lambda/2$.

Strong duality!

$p^* = d^*$

$$q(\lambda) = \lambda - \frac{\lambda^2}{4}. \Leftarrow \left(\frac{\lambda}{2}\right)^2 + \lambda\left(1 - \frac{\lambda}{2}\right)$$

Dual problem:

$d^* = 1$

$$(D) \quad \max_{\lambda \geq 0}; \lambda - \frac{\lambda^2}{4}.$$

$1 - \frac{2\lambda}{4} = 0$

$\lambda = 2$

Duality and Complementary Slackness

We are given the linear program:

$3(0) + 2(4.5) = 9$

$p^* = 9$

$$\max_{x_1, x_2, x_3 \geq 0} \quad 3x_1 + 2x_3$$

$$\text{s.t.} \quad \begin{cases} x_1 + x_2 + x_3 \leq 6 \\ 2x_1 - x_2 + x_3 \leq 3 \\ 3x_1 + x_2 - x_3 \leq 3 \\ x_1, x_2, x_3 \geq 0 \end{cases}$$

$c^T x$

$Ax \leq b$

$g_j(x) \leq 0$

$(x_1 + x_2 + x_3 - 6) = 0$

$x_1 = 0$

$x_2 + x_3 = 6$

$-x_2 + x_3 = 3$

$x_3 = 4.5$

$x_2 = 1.5$

A candidate optimal solution is $(x_1, x_2, x_3) = (0, 1.5, 4.5)$.

one of the intersection pts in the feasible region.

We first verify feasibility:

- $x_1, x_2, x_3 \geq 0$ ✓
- $x_1 + x_2 + x_3 = 6$ ✓
- $2x_1 - x_2 + x_3 = 3$ ✓
- $3x_1 + x_2 - x_3 = -3 < 3$ ✓

$2x_1 - x_2 + x_3 - 3 = 0$

at x^*

Thus, the point is feasible and a valid candidate for optimality.

$$x_1 + x_2 + x_3 \le 6$$
$$2x_1 - x_2 + x_3 \le 3$$
$$3x_1 + x_2 - x_3 \le 3$$

$$\begin{cases} x_1 + x_2 + x_3 - 6 \le 0 \\ 2x_1 - x_2 + x_3 - 3 \le 0 \\ 3x_1 + x_2 - x_3 - 3 \le 0 \end{cases}$$

$g_j(x) \le 0$

at $x^* = (0, 1.5, 4.5)$

1) $0 + 1.5 + 4.5 - 6 = 0$ — tight
2) $3 - 3 = 0$ — tight
3) $-3 - 3 \ne 0$ — slack

$u_i^* \cancel{g_i(x^*)} = 0$ for 1st & 2nd

$u_3 = 0$.

Dual Problem

The dual of the primal LP is: $(1, 1, 0)$ $1 + 2 + 0$

$d^* = 9$ $\Rightarrow$

$$\min_{u_1, u_2, u_3 \geq 0} \quad 6u_1 + 3u_2 + 3u_3$$ $b^T u$

$$\text{s.t.} \begin{cases} u_1 + 2u_2 + 3u_3 \geq 3 \\ u_1 - u_2 + u_3 \geq 0 \\ u_1 + u_2 - u_3 \geq 2 \end{cases}$$

$g_j(x^*) \leq 0$ $A^T u \geq c$

2 primal feasibility $u \geq 0$

$(0, 1.5, 4.5)$

At the primal point:

- Constraint 1: $x_1 + x_2 + x_3 = 6$ (tight)
- Constraint 2: $2x_1 - x_2 + x_3 = 3$ (tight)
- Constraint 3: $3x_1 + x_2 - x_3 < 3$ (slack)

$u_3(g_3(x)) = 0$

$u_3 = 0$

Since the last constraint is strictly slack (not tight), complementary slackness implies the corresponding dual variable must be zero:

$$u_3\big(3 - (3x_1 + x_2 - x_3)\big) = 0 \implies \boxed{u_3 = 0}.$$

Complementary Slackness: Dual Constraints

Since the primal variables $x_2 = 1.5 > 0$ and $x_3 = 4.5 > 0$, the corresponding dual constraints (D2) and (D3) must be binding (tight):

$$u_1 - u_2 + \cancel{u_3} = 0,$$
$$u_1 + u_2 - \cancel{u_3} = 2.$$

Substituting $u_3 = 0$ into this system yields:

$$\begin{cases} u_1 - u_2 = 0 \\ u_1 + u_2 = 2 \end{cases}$$

Solving this simple system gives:

$$2u_1 = 2 \implies u_1 = 1, \quad \text{and} \quad u_2 = 1.$$

Thus, the candidate dual solution is:

$$\boxed{(u_1, u_2, u_3) = (1, 1, 0)}.$$

Dual Feasibility and Optimality

(1, 1, 0) is dual feasible!

We must verify if $(1, 1, 0)$ satisfies the remaining dual constraint :

$$u_1 + 2u_2 + 3u_3 = 1 + 2(1) + 3(0) = 3 \geq 3.$$

The constraint holds. Since $(1, 1, 0)$ is dual feasible and satisfies complementary slackness with the feasible $\bar{x}$, both are optimal. The common optimal objective value is:

(0, 1.5, 4.5) is optimal!

$$3(0) + 2(4.5) = 9.$$

$d^* = 9$

$p^* \leq d^*$

$p^* \leq 9$

Characterization of All Primal Optimal Solutions

To find *all* optimal primal solutions, we note that the optimal dual variables are strictly positive ($u_1 > 0$ and $u_2 > 0$). By complementary slackness, this forces the first two primal constraints to be tight for *any* optimal solution:

$$\begin{cases} x_1 + x_2 + x_3 = 6, & (1) \\ 2x_1 - x_2 + x_3 = 3. & (2) \end{cases}$$

$x_1 = t$
$x_2 = 1.5 + 0.5t$
$x_3 = 4.5 - 1.5t$

Let $x_1 = t$ be the parameter. Subtracting the equations eliminates x_3 temporarily, or we can solve generally. From equation 1: $x_2 + x_3 = 6 - t$. From equation 2: $-x_2 + x_3 = 3 - 2t$. Adding these two:

$$2x_3 = 9 - 3t \implies x_3 = 4.5 - 1.5t.$$

Substituting back to find x_2:

$$x_2 = (6 - t) - (4.5 - 1.5t) = 1.5 + 0.5t.$$

So the solutions are parametrized as:

$$(x_1, x_2, x_3) = (t,\ 1.5 + 0.5t,\ 4.5 - 1.5t).$$

primal

Feasibility Conditions

For this solution to be valid, it must satisfy non-negativity ($x \geq 0$) and the third constraint (P3).

$x_1, x_2, x_3 \geqslant 0$

dual.

KKT

1. **Non-negativity:**

$$t \geq 0$$
$$1.5 + 0.5t \geq 0 \implies t \geq -3$$
$$4.5 - 1.5t \geq 0 \implies 1.5t \leq 4.5 \implies t \leq 3$$

   So currently $0 \leq t \leq 3$.

2. **Constraint (P3):**

$$
\begin{aligned}
3x_1 + x_2 - x_3 &\leq 3 \\
3(t) + (1.5 + 0.5t) - (4.5 - 1.5t) &\leq 3 \\
3t + 1.5 + 0.5t - 4.5 + 1.5t &\leq 3 \\
5t - 3 &\leq 3 \\
5t &\leq 6 \implies t \leq 1.2
\end{aligned}
$$

Combining the conditions, the valid range for t is $0 \leq t \leq 1.2$.

Conclusion

$3x_1 + 2x_3$

$3t + 2(4.5 - 1.5t) = 9$

1. **Primal Solution :** The primal problem gave us the solution set:

$$\{(t,\ 1.5 + 0.5t,\ 4.5 - 1.5t) \mid 0 \le t \le 1.2\}.$$

Despite the multiple x^* , the maximum objective value is unique and constant at **9**.

2. **Dual Solution:** The dual problem provided us with the solution:

$$(u_1^*, u_2^*, u_3^*) = (1, 1, 0).$$

$d^* = 9$

The minimum dual cost is **9**, matching the primal solution.

Role of Duality in Solving the Problem

Duality plays a fundamental role in solving the linear program by providing both a *global bound* on the objective value and a *certificate of optimality*. These are guarantees that the primal formulation alone does not directly provide.

By weak duality, for any primal feasible x and any dual feasible u, the following inequality holds:

$$3x_1 + 2x_3 \;\leq\; 6u_1 + 3u_2 + 3u_3.$$

Thus, every dual feasible point produces an *upper bound* on the primal objective. In particular, the dual feasible point

$$(u_1, u_2, u_3) = (1, 1, 0)$$

$p^* \leq d^*$

yields

$$3x_1 + 2x_3 \leq 6(1) + 3(1) + 3(0) = 9$$

for *all* primal feasible x.

Conclusion

Therefore, duality proves that no feasible primal solution can have objective value exceeding 9.

Solving the primal problem directly would typically require enumerating vertices of the feasible polytope or executing the simplex algorithm.

Duality avoids this entirely. Specifically, duality:

- provides a *global optimality certificate* without vertex enumeration,
- explains *why* the objective cannot be increased further,
- identifies which constraints are binding through complementary slackness,
- characterizes *all* optimal solutions, not just one.

any candidate primal probs!

Conclusion

Therefore, duality proves that no feasible primal solution can have objective value exceeding 9.

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- provides a *global optimality certificate* without vertex enumeration,
- explains *why* the objective cannot be increased further,
- identifies which constraints are binding through complementary slackness,
- characterizes *all* optimal solutions, not just one.

$\max \; 7x_1 + 6x_2$

s.t. $2x_1 + 4x_2 \le 16$

$3x_1 + 2x_2 \le 12$

$x_1, x_2 \ge 0$

$Z = 7x_1 + 6x_2 + 0s_1 + 0s_2$

$2x_1 + 4x_2 + s_1 = 16$

$3x_1 + 2x_2 + s_2 = 12$

$x_1 = x_2 = 0$, $s_1 = 16$, $s_2 = 12$

①

C_B		7	6	0	0		
		x_1	x_2	s_1	s_2	RHS	
0	s_1	2	4	1	0	16	$2 = 8$
0	s_2	3	2	0	1	12	$3 = 4$
	Z_j	0	0	0	0	0	
	$C_j - Z_j$	7	6	0	0		

min: most negative

max: most positive

ratio: lowest one leaves

gaussian elimination.

① $R_2 \to R_2/3$

② $R_1 \to R_1 - 2R_2$

②

			x_2				
0	s_1	0	$8/3$	1	$-2/3$	8	$/ (8/3) = 3$
7	x_1	1	$2/3$	0	$1/3$	4	$/ (2/3) = 6$
	Z_j	7	$14/3$	0	$7/3$	28	
	$C_j - Z_j$	0	$4/3$	0	$-7/3$		

$0(0) + 7(1) : Z_1$

$0(8/3) + 7(2/3) : Z_2$

⋮

3)
$$\begin{array}{c|c|cccc|c}
6 & x_2 & 0 & 1 & 3/8 & -1/4 & 3 \\
7 & x_1 & 1 & 0 & -1/4 & 1/2 & 2 \\
\hline
 & Z_j & 7 & 6 & 1/2 & 2 & 32 \\
\hline
 & C_j - Z_j & 0 & 0 & -1/2 & -2 &
\end{array}$$

① $R_1 \to R_1(3/8)$

② $R_2 \to R_2 - \frac{2}{3}R_1$

Z_j:
$Z_1 : 6(0) + 7(1)$
$Z_2 : 6(1) + 7(0)$
$6(3/8) + 7(-1/4)$
$6(-1/4) + 7(1/2)$

max all positive
STOP!

RHS: $6(3) + 7(2)$
$= 32$

$$\Rightarrow \begin{array}{c} x_1 \\ x_2 \end{array} \left[\begin{array}{cc|cc|c} 1 & 0 & -1/4 & 1/2 & 2 \\ 0 & 1 & 3/8 & -1/4 & 3 \end{array}\right]$$

$[\, I \mid A^{-1} \,]$

$A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$

$A^{-1} = -\frac{1}{8} \begin{bmatrix} 2 & -4 \\ -3 & 2 \end{bmatrix}$

at $(x_1, x_2) = (2, 3)$, $P^* = 32$

$\max f = 7x_1 + 6x_2 \Rightarrow \begin{cases} 2x_1 + 4x_2 \le 16 \\ 3x_1 + 2x_2 \le 12 \end{cases}$